Systems Analysis and Control

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Lecture 5: Calculating the Laplace Transform of a Signal

Introduction

In this Lecture, you will learn:

Laplace Transform of Simple Signals.

- Step, Exponentials, Ramps.
- Impulse

Properties of the Laplace Transform.

- delay, time-scaling
- convolution
- etc.

Previously: The Laplace Transform

Definition 1.

The **Laplace Transform** of the signal $u(t)$ is

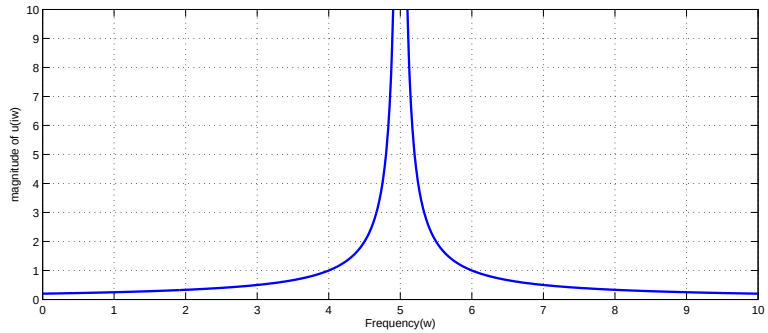
$$\hat{u}(s) = \int_0^\infty e^{-st} u(t) dt$$

The Laplace Transform is not the same as the Fourier transform.

- Assumes that signals begin at time $t = 0$ and $u(t) = 0$ for $t < 0$.
- Tends to flatten out peaks.

Signal:

$$u(t) = \sin at$$



Laplace Transform:

$$\hat{u}(s) = \frac{a}{s^2 + a^2}$$

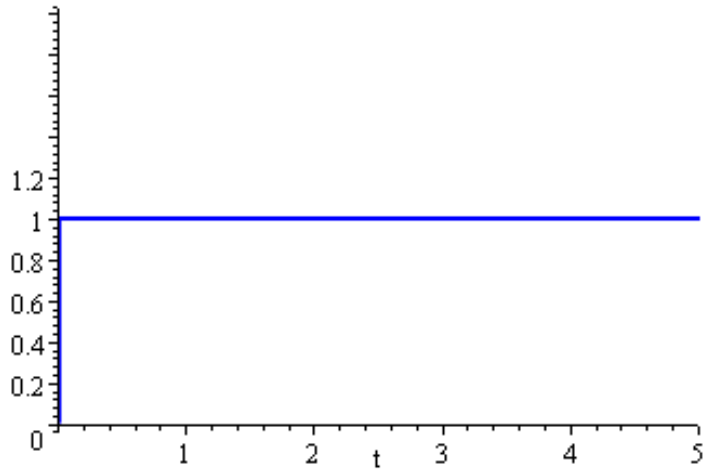
Recall: The Fourier Transform of $u(t)$ was $\delta(\omega - a)$.

The Laplace Transform

Example: Step Function

Consider the input signal:

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

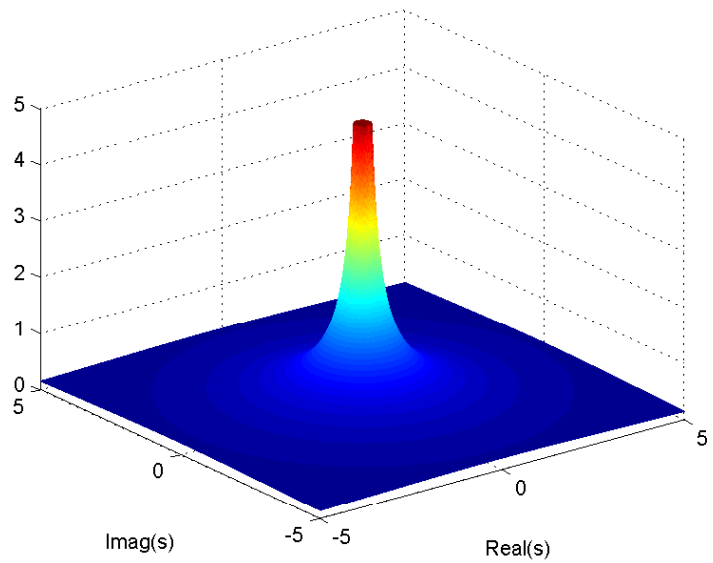


The Laplace transform is

$$\begin{aligned} \hat{u}(s) &= \int_0^\infty e^{-st} dt \\ &= \frac{1}{-s} \left[e^{-st} \right]_0^\infty = \frac{1}{-s} [0 - 1] \\ &= \frac{1}{s} \end{aligned}$$

Note that $\lim_{t \to \infty} e^{-st} = 0$.

- Because we assume the real part of s is negative.



Review: Integration by parts

A very useful formula for finding the Laplace Transform is

- integration by parts.

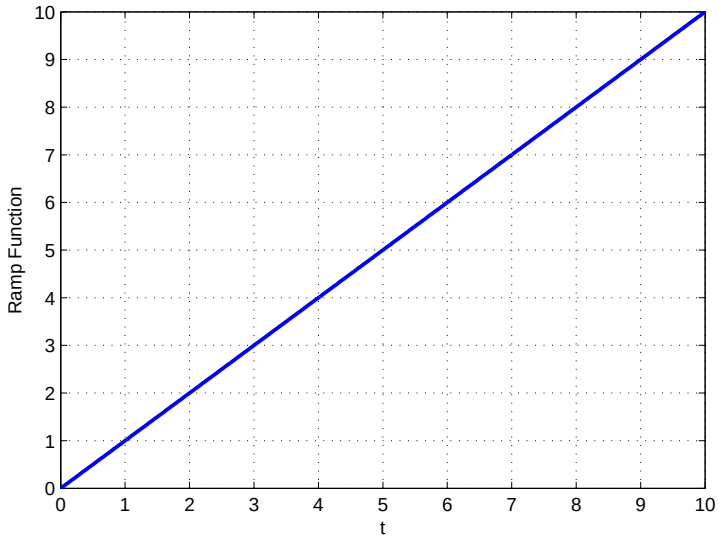
$$\int_a^b u(s)v'(s)ds = [u(s)v(s)]_{s=a}^{s=b} - \int_a^b u'(s)v(s)ds$$

The Laplace Transform

Example: Ramp Function

Consider the input signal:

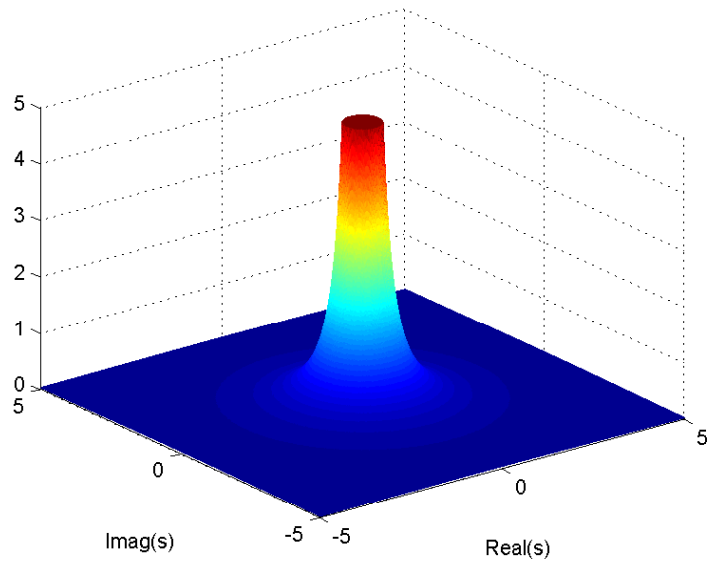
$$u(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases}$$
$$= t \cdot \mathbf{1}(t)$$



Recall $\mathbf{1}(t)$ is the step function.

The Laplace transform is

$$\hat{u}(s) = \int_0^\infty te^{-st}dt$$
$$= \left[\frac{1}{-s}te^{-st}\right]_0^\infty + \frac{1}{s}\int_0^\infty e^{-st}dt$$
$$= [0-0] - \frac{1}{s^2}[e^{-st}]_0^\infty dt$$
$$= -\frac{1}{s^2}[0-1] = \frac{1}{s^2}$$



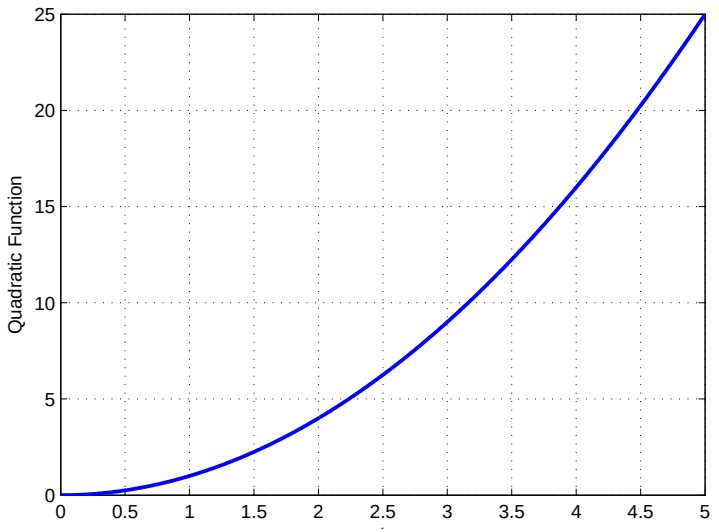
The Laplace Transform

Example: Quadratic Function

The Quadratic is similar to the Ramp.

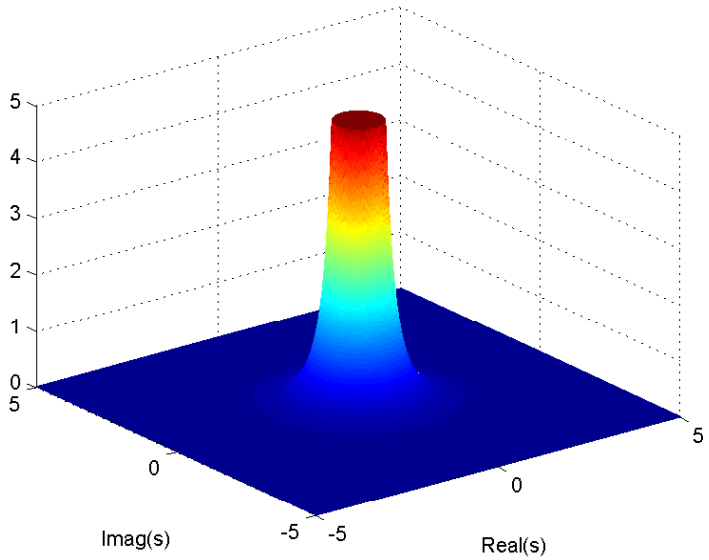
$$u(t) = \begin{cases} 0 & t < 0 \\ t^2 & t \geq 0 \end{cases}$$
$$= t^2 \mathbf{1}(t)$$

Recall $\mathbf{1}(t)$ is the step function.



The Laplace transform is

$$\hat{u}(s) = \int_0^\infty t^2 e^{-st} dt$$
$$= \left[\frac{1}{-s} t^2 e^{-st} \right]_0^\infty + \frac{1}{s} \int_0^\infty t e^{-st} dt$$
$$= \frac{1}{s^3}$$



The Laplace Transform

Example: exponential

The Sinusoid is created from the complex exponentials.

- Lets do exponentials first.

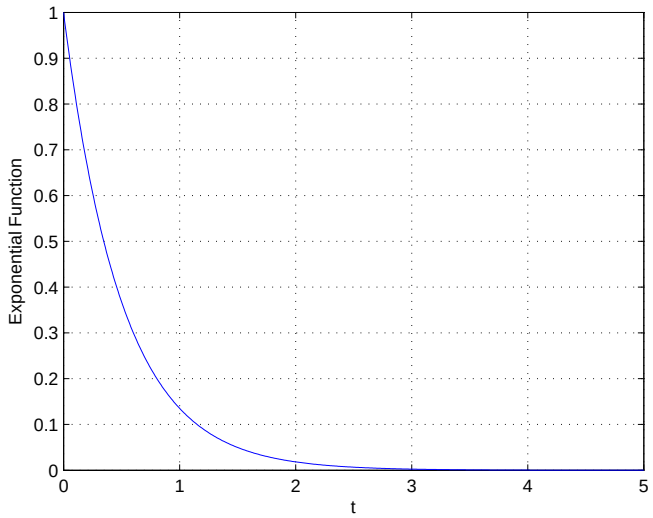
Consider the input signal:

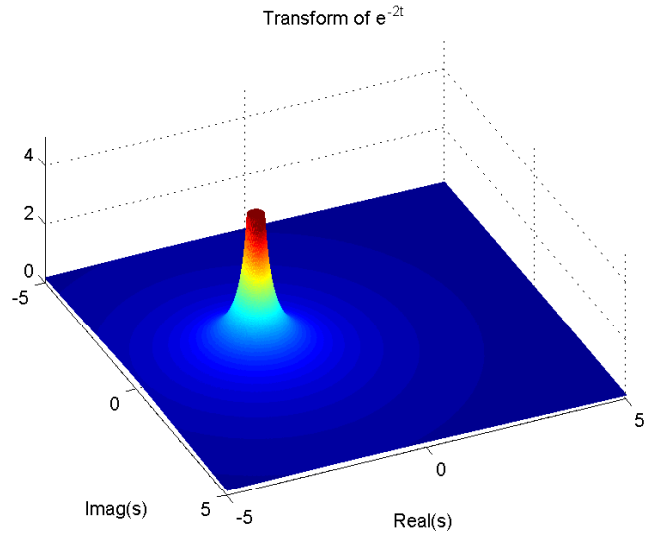
$$u(t) = \begin{cases} 0 & t < 0 \\ e^{at} & t \geq 0 \end{cases}$$
$$= e^{at}\mathbf{1}(t)$$

Recall $\mathbf{1}(t)$ is the step function.

The Laplace transform is

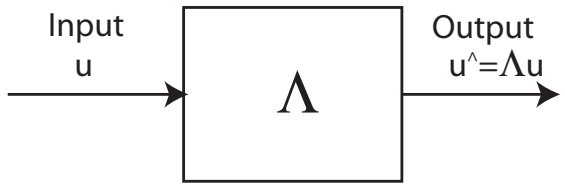
$$\hat{u}(s) = \int_0^\infty e^{(a-s)t} dt$$
$$= \left[\frac{1}{a-s} e^{(a-s)t} \right]_0^\infty$$
$$= \frac{1}{s-a}$$





The Laplace Transform

Linearity Property



The Laplace Transform is itself a LINEAR SYSTEM.

- The *Input* is u
- The *Output* is $\hat{u}$
- Lets call the system $\mathcal{L}$

$$\hat{u} = \mathcal{L}u$$

- As for any linear system,

$$\begin{aligned}\mathcal{L}(\alpha u_1 + \beta u_2) &= \alpha \mathcal{L} u_1 + \beta \mathcal{L} u_2 \\ &= \alpha \hat{u}_1 + \beta \hat{u}_2\end{aligned}$$

We can compute the Laplace transform by using simple parts.
Example: Sinusoids

$$\cos \omega t = \frac{e^{\imath \omega t} + e^{-\imath \omega t}}{2}$$

The Laplace Transform

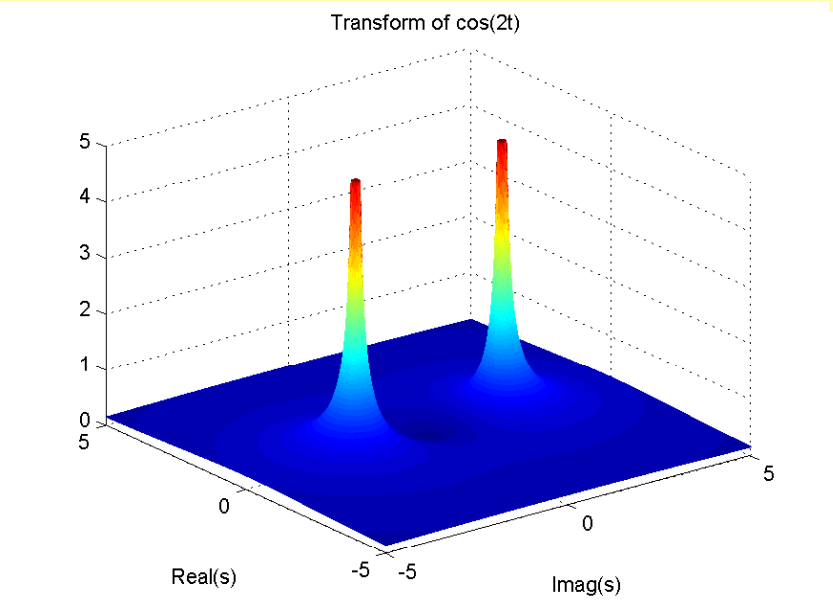
Example: Sinusoids

Recall that

$$\cos \omega t = \frac{e^{\imath \omega t} + e^{-\imath \omega t}}{2}$$

But the Laplace transform of e^{at} is

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}$$



So the Laplace transforms of $e^{\imath \omega t}$ and $e^{-\imath \omega t}$ are

$$\hat{u}_1(s) = \mathcal{L}(e^{\imath \omega t}) = \frac{1}{s - \imath \omega} \qquad \text{and} \qquad \hat{u}_2(s) = \mathcal{L}(e^{-\imath \omega t}) = \frac{1}{s + \imath \omega}$$

Which shows us how to find $\mathcal{L}(\cos(\omega t))$

$$\mathcal{L}(\cos(\omega t)) = \mathcal{L}\left(\frac{e^{\imath \omega t} + e^{-\imath \omega t}}{2}\right) = \mathcal{L}\left(\frac{e^{\imath \omega t}}{2}\right) + \mathcal{L}\left(\frac{e^{-\imath \omega t}}{2}\right) = \frac{\hat{u}_1}{2} + \frac{\hat{u}_2}{2}$$

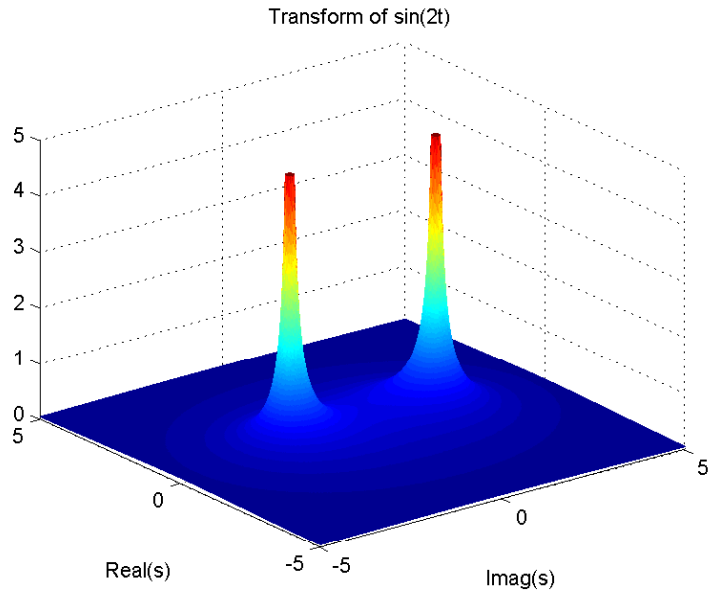
$$= \frac{1}{2}\left(\frac{1}{s - \imath \omega} + \frac{1}{s + \imath \omega}\right) = \frac{1}{2}\left(\frac{2s}{(s - \imath \omega)(s + \imath \omega)}\right) = \frac{s}{s^2 + \omega^2}$$

The Laplace Transform

Example: Sinusoids

Likewise, for the $\sin$ function,

$$\begin{aligned}\sin \omega t &= \frac{e^{\imath\omega t} - e^{-\imath\omega t}}{2\imath} \\ &= \frac{1}{2\imath}\left(u_1(t) - u_2(t)\right) \\ u_1(t) &= e^{\imath\omega t} \\ u_2(t) &= e^{-\imath\omega t}\end{aligned}$$



Which shows us how to find $\mathcal{L}(\sin(\omega t))$

$$\begin{aligned}\mathcal{L}(\sin(\omega t)) &= \frac{1}{2\imath}\left(\hat{u}_1(s) - \hat{u}_2(s)\right) \\ &= \frac{1}{2\imath}\left(\frac{1}{s - \imath\omega} - \frac{1}{s + \imath\omega}\right) \\ &= \frac{1}{2\imath}\left(\frac{2\imath\omega}{(s - \imath\omega)(s + \imath\omega)}\right) = \frac{\omega}{s^2 + \omega^2}\end{aligned}$$

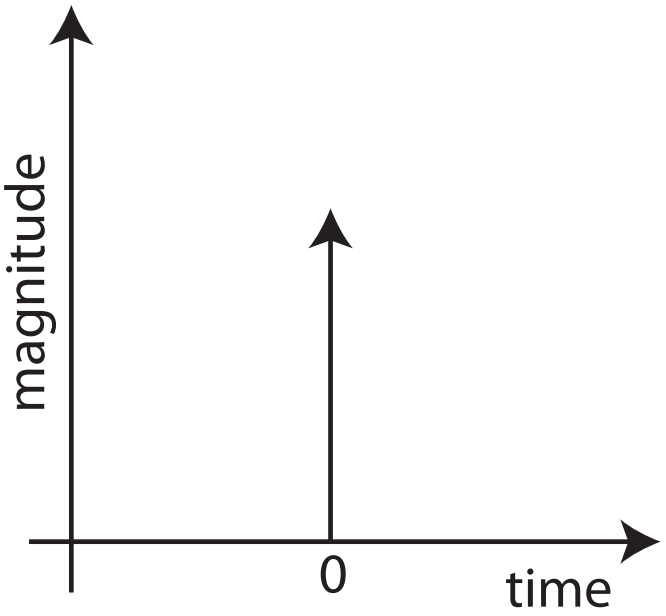
Definition 2.

An **Impulse** is a significant strike which occurs in an infinitesimally short time. Mathematically, we model the impulse as a Dirac delta function, $\delta(t)$.

Question: What is the Laplace transform of an *Impulse*?

Answer: By definition of the Dirac Delta:

$$\begin{aligned}\hat{\delta}(s) &= \int_0^\infty e^{-st}\delta(t)dt \\ &= e^{-st}|_{t=0} \\ &= 1\end{aligned}$$



- The Impulse has a Uniform Frequency Response!!!!
- The opposite of the step function ($\mathbf{1}(s) = \mathcal{L}\delta(t)$).

Laplace Transform Table

Simple Signals

Summary: We can find the Laplace Transform for many simple signals

	$u(t)$	$\hat{u}(s)$
Step	$\mathbf{1}(t)$	$\frac{1}{s}$
Power	t^m	$\frac{m!}{s^{m+1}}$
Exponential	e^{-at}	$\frac{1}{s+a}$
Power Exponential	$\frac{t^{m-1}e^{-at}}{(m-1)!}$	$\frac{1}{(s+a)^m}$
Sine	$\sin(at)$	$\frac{a}{s^2+a^2}$
Cosine	$\cos(at)$	$\frac{s}{s^2+a^2}$
Impulse	$\delta(t)$	1

Remember that all functions are only for $t \geq 0$.

- We always have $u(t) = 0$ for $t < 0$

Problem: Computing $\hat{u}(s)$ manually is Boring.
Solution: Only use Variations and Combinations of Simple Functions.

Properties of the Laplace Transform

Properties: What properties of the Laplace transform can we exploit?

- We have already discussed *Linearity*.

Question: Suppose we know the Laplace Transform of $u(t)$ is $\hat{u}(s)$. What are some simple variations of $u(t)$?

Simple Variation:

- **Delay:** The signal is delayed

$$u(t) \to u(t-\tau) \qquad \hat{u}(s) \to e^{-s\tau}\hat{u}(s)$$

- **Time-Scale:** The signal is slowed down or speeded up

$$u(t) \to u(at) \qquad \hat{u}(s) \to \frac{1}{a}\hat{u}\left(\frac{s}{a}\right)$$

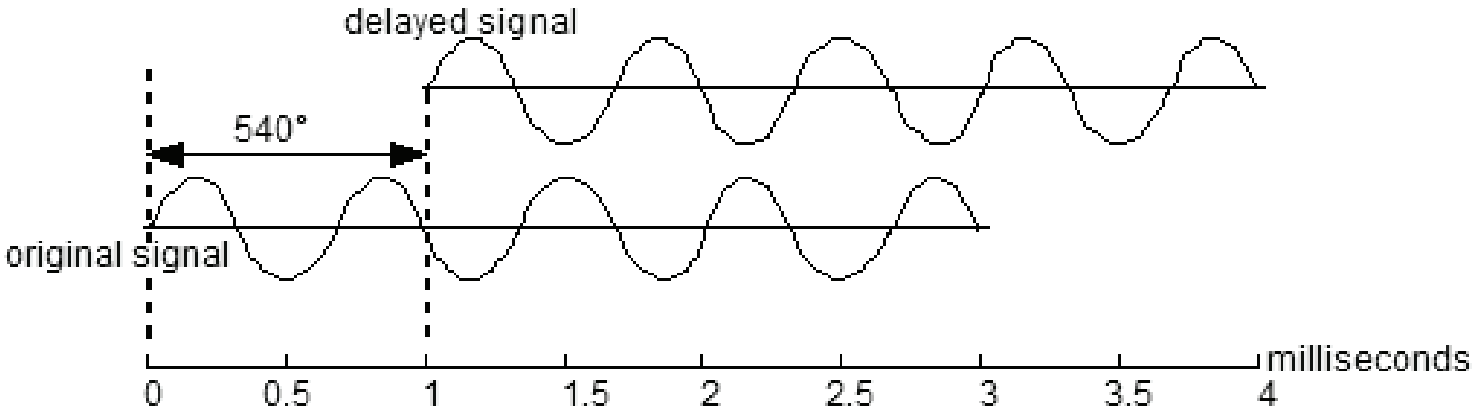
- **Differentiation:** The signal is a differential in time:

$$u(t) \to u'(t) \qquad \hat{u}(s) \to s\hat{u}(s)$$

How do these changes affect the the Laplace transform?

Effect of Delay on the Laplace Transform

What is the effect of a delay on the Laplace Transform?



Use a change of variables: $t' = t - \tau$, $dt' = dt$

$$\hat{u}_{delayed}(s) = \mathcal{L}(u(t-\tau)) = \int_0^\infty e^{-st} u(t-\tau) dt$$

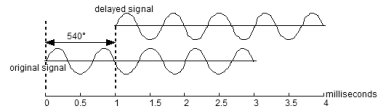
$$= \int_0^\infty e^{-s(t'+\tau)} u(t') dt'$$

$$= e^{-s\tau} \int_0^\infty e^{-st'} u(t') dt' = e^{-s\tau} \hat{u}(s)$$

$$u(t) \to u(t-\tau) \qquad \hat{u}(s) \to e^{-s\tau} \hat{u}(s)$$

Effect of Delay on the Laplace Transform

What is the effect of a delay on the Laplace Transform?



Use a change of variables: $t' = t - \tau$, $dt' = dt$

$$\hat{u}_{delayed}(s) = \mathcal{L}(u(t-\tau)) = \int_0^\infty e^{-st}u(t-\tau)dt$$
$$= \int_0^\infty e^{-s(t'+\tau)}u(t')dt'$$
$$= e^{-s\tau}\int_0^\infty e^{-st'}u(t')dt' = e^{-s\tau}\hat{u}(s)$$

$u(t) \to u(t-\tau)$ $\quad$ $\hat{u}(s) \to e^{-s\tau}\hat{u}(s)$

Example 1: Recall that for $u(t) = \sin(at)$, we have $\hat{u}(s) = \frac{a}{s^2+a^2}$.

$$u(t) \to u(t-\tau) \qquad \hat{u}(s) \to e^{-s\tau}\hat{u}(s)$$

Conclusion: If $u(t) = \sin(a(t-\tau))$, we have $\hat{u}(s) = e^{-s\tau}\frac{a}{s^2+a^2}$.

Example 2: Recall that for $u(t) = t^2$, we have $\hat{u}(s) = \frac{2}{s^3}$.

$$u(t) \to u(t-\tau) \qquad \hat{u}(s) \to e^{-s\tau}\hat{u}(s)$$

Conclusion: If $u(t) = (t-\tau)^2$, we have $\hat{u}(s) = \frac{2e^{-s\tau}}{s^3}$.

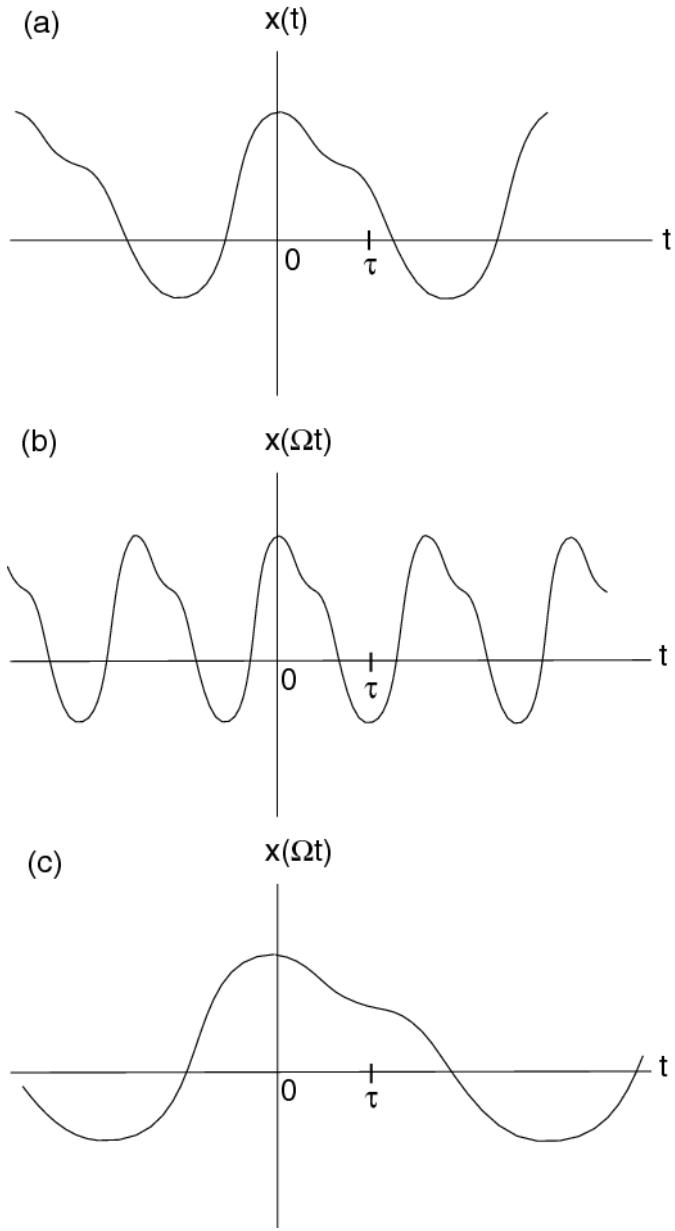
Effect of Time-Scaling on the Laplace Transform

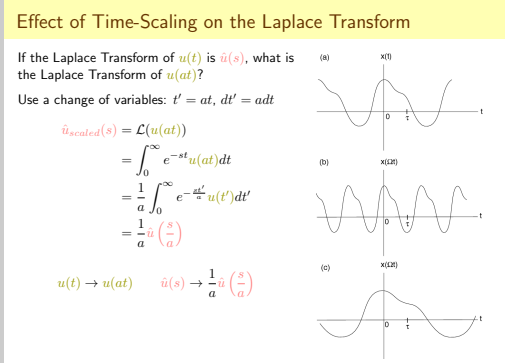
If the Laplace Transform of $u(t)$ is $\hat{u}(s)$, what is the Laplace Transform of $u(at)$?

Use a change of variables: $t' = at$, $dt' = adt$

$$\begin{aligned}\hat{u}_{scaled}(s) &= \mathcal{L}(u(at)) \\ &= \int_0^\infty e^{-st} u(at) dt \\ &= \frac{1}{a}\int_0^\infty e^{-\frac{st'}{a}} u(t') dt' \\ &= \frac{1}{a}\hat{u}\left(\frac{s}{a}\right)\end{aligned}$$

$$u(t) \to u(at) \qquad \hat{u}(s) \to \frac{1}{a}\hat{u}\left(\frac{s}{a}\right)$$





Example 1: Recall that for $u(t) = \cos(t)$, we have $\hat{u}(s) = \frac{s}{s^2+1}$.

$$u(t) \to u(at) \qquad \hat{u}(s) \to \frac{1}{a}\hat{u}\left(\frac{s}{a}\right)$$

Conclusion: If $u(t) = \cos(at)$, we have

$$\hat{u}(s) = \frac{1}{a}\frac{\frac{s}{a}}{\left(\frac{s}{a}\right)^2 + 1} = \frac{s}{s^2 + a^2}.$$

Of course, we could have just used the Table!

Example 2: Recall that for $u(t) = t^2$, we have $\hat{u}(s) = \frac{2}{s^3}$.

$$u(t) \to u(at) \qquad \hat{u}(s) \to \frac{1}{a}\hat{u}\left(\frac{s}{a}\right)$$

Conclusion: If $u(t) = at^2 = (\sqrt{a}t)^2$, we have

$$\hat{u}(s) = \frac{1}{\sqrt{a}}\frac{2}{(\frac{s}{\sqrt{a}})^3} = \frac{\sqrt{a}^3}{\sqrt{a}}\frac{2}{s^3} = \frac{2a}{s^3}.$$

Of course, we could have just used linearity of the Laplace Transform!

Example: Watch The Order of Operations

What is the Laplace Transform of

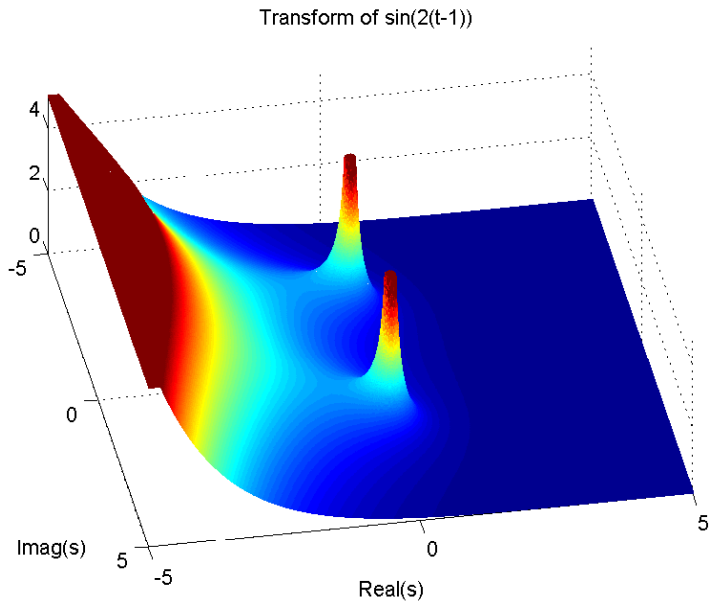
$$u(t) = \sin(at - \tau)?$$

We know that $\hat{u}_{\sin}(s) = \mathcal{L}(\sin(t)) = \frac{1}{s^2+1}$.

- The change is $\sin t \to \sin(t-\tau) \to \sin(at-\tau)$
- First **Time-Delay**: $\hat{u}_d(s) = e^{-\tau s}\hat{u}(s)$
- Second **Time-Scaling**: $\hat{u}_s(s) = \frac{1}{a}\hat{u}_d\left(\frac{s}{a}\right)$

Thus we have

$$\begin{aligned}\hat{u}_{ds}(s) &= \frac{1}{a}\hat{u}_d\left(\frac{s}{a}\right) \\ &= \frac{1}{a}\left(\frac{e^{-\tau\frac{s}{a}}}{\left(\frac{s}{a}\right)^2+1}\right) \\ &= e^{-\frac{\tau}{a}s}\frac{a}{s^2+a^2}\end{aligned}$$



The Laplace Transform Properties

Differentiation

A very common case is when a signal has been differentiated. e.g.:

$$\dot{x}(t) = Ax(t)$$

Suppose $\hat{x}(s) = \mathcal{L}(x(t))$. What is $\mathcal{L}(\dot{x}(t))$?

In this case we use integration by parts:

$$\begin{aligned}
\mathcal{L}\left(\dot{x}(t)\right) &= \int_0^\infty e^{-st}\dot{x}(t)dt \\
&= \left[e^{-st}x(t)\right]_0^\infty + s\int_0^\infty e^{-st}x(t)dt \\
&= -x(0) + s\int_0^\infty e^{-st}x(t)dt \\
&= -x(0) + s\hat{x}(s)
\end{aligned}$$

Notice the $x(0)$! This is why we usually assume $x(0) = 0$ for all our systems.

The differentiation property can be applied recursively:

$$\begin{aligned}\mathcal{L}\left(\dddot{x}(t)\right) &= -\ddot{x}(0) + s\mathcal{L}(\ddot{x}(t)) \\ &= -\ddot{x}(0) - s\dot{x}(0) + s^2\mathcal{L}(\dot{x}(t)) \\ &= -\ddot{x}(0) - s\dot{x}(0) - s^2x(0) + s^3\hat{x}(s)\end{aligned}$$

General Formula:

$$\mathcal{L}(x^{(n)}(t)) = s^n\hat{x}(s) - s^{n-1}x(0) - \cdots - x^{n-1}(0)$$

The Laplace Transform Properties

Integration

Importantly, the Laplace transform can also be applied to integration:

- An immediate consequence of differentiation

$$\mathcal{L}\left(\dot{f}(t)\right) = -f(0) + s\mathcal{L}(f(t))$$

- If $f(t) = \int_0^t x(\tau)d\tau$, then $\dot{f}(t) = x(t)$, and $f(0) = 0$, so

$$\begin{aligned}\hat{x}(s) = \mathcal{L}(x(t)) &= \mathcal{L}\left(\dot{f}(t)\right) \\ &= -f(0) + s\mathcal{L}(f(t)) \\ &= s\mathcal{L}\left(\int_0^t x(\tau)d\tau\right)\end{aligned}$$

- Hence

$$\mathcal{L}\left(\int_0^t x(\tau)d\tau\right) = \frac{\hat{x}(s)}{s}$$

Recall the solution of a state-space system:

$$y(t) = \int_0^t Ce^{A(t-s)}Bu(s)ds + Du(t)$$

This is a special kind of integral called the *Convolution Integral*.

Definition 3.

The **Convolution Integral** of $u(t)$ with $v(t)$ is defined as

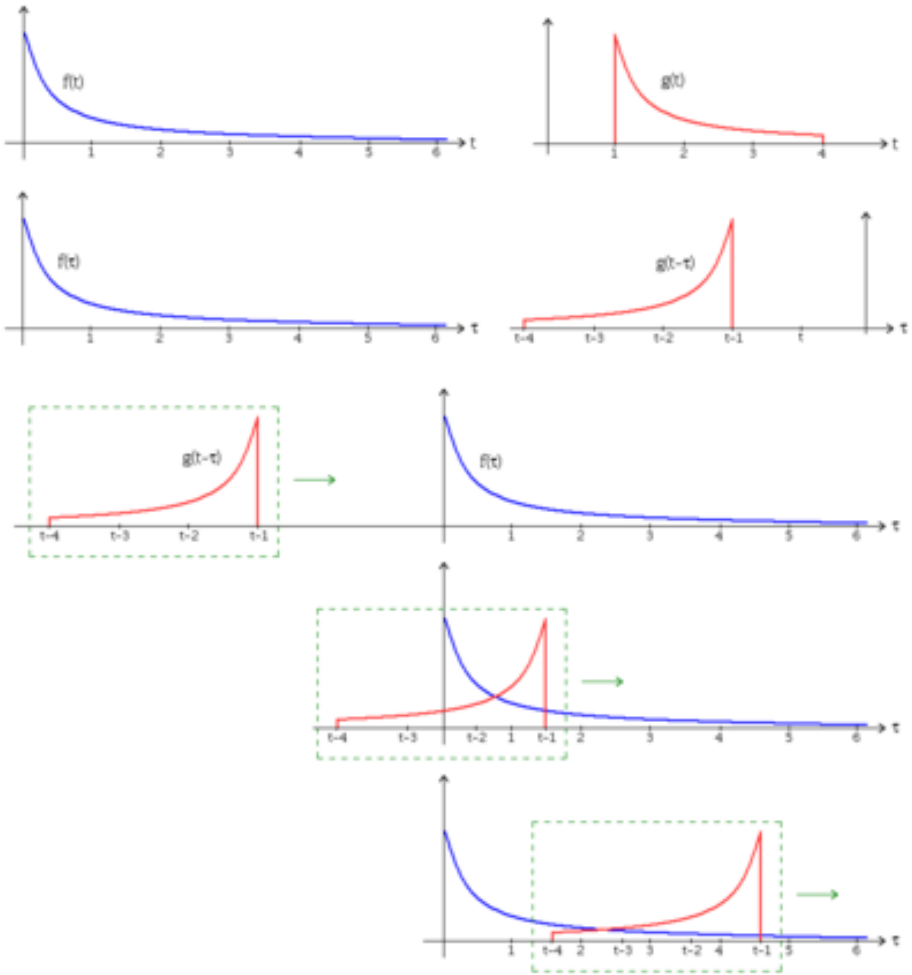
$$(u * v)(t) := \int_0^t u(\tau)v(t-\tau)d\tau$$

Thus roughly speaking

$$y(t) = u(t) * \left(Ce^{At}B\right)$$

The Laplace Transform Properties

Convolution



$$(f * g)(t) := \int_0^t f(\tau)g(t-\tau)d\tau$$

The Laplace Transform Properties

Convolution

The convolution has a remarkable property: if $\hat{u}(s) = \mathcal{L}(u(t))$ and $\hat{v}(s) = \mathcal{L}(v(t))$, then

$$\mathcal{L}(u * v) = \hat{u}(s)\hat{v}(s)$$

Proof.

Since $u(t) = v(t) = 0$ for $t < 0$, we have that $u(\tau)v(t-\tau) = 0$ for $\tau < 0$ or $\tau > t$. Thus:

$$\int_0^t u(\tau)v(t-\tau)d\tau = \int_0^\infty u(\tau)v(t-\tau)d\tau$$

Now using the delay property:

$$\begin{aligned}
\mathcal{L}(u * v) &= \int_0^\infty e^{-st} \int_0^t u(\tau)v(t-\tau)d\tau dt = \int_0^\infty e^{-st} \int_0^\infty u(\tau)v(t-\tau)d\tau dt \\
&= \int_0^\infty u(\tau) \int_0^\infty e^{-st} v(t-\tau)dt d\tau \\
&= \int_0^\infty u(\tau)e^{-s\tau}\hat{v}(s)d\tau = \hat{u}(s)\hat{v}(s)
\end{aligned}$$



Properties of the Laplace Transform

A Table of Useful Properties

Summary: Simple functions and properties can be combined to calculate the Laplace Transform.

	$u(t)$	$\hat{u}(s)$
Delay	$u(t-\tau)$	$e^{-\tau s}\hat{u}(s)$
Time-Scaling	$u(at)$	$\frac{1}{a}\hat{u}\left(\frac{s}{a}\right)$
Differentiation	$u'(t)$	$s\hat{u}(s)-u(0)$
Integration	$\int_0^t u(\tau)d\tau$	$\frac{1}{s}\hat{u}(s)$
Frequency Differentiation	$tu(t)$	$\frac{d}{ds}\hat{u}(s)$
Frequency Shift	$e^{at}u(t)$	$\hat{u}(s-a)$
Convolution	$\int_0^t u(\tau)v(t-\tau)d\tau$	$\hat{u}(s)\hat{v}(s)$

The Inverse Laplace Transform

Tables can be used to calculate the **Inverse** Laplace transform of a signal.
Question: Given $\hat{u}(s)$, what is $u(t) = \mathcal{L}^{-1}(\hat{u}(s))$?

	$u(t)$	$\hat{u}(s)$
Step	$1(t)$	$\frac{1}{s}$
Power	t^m	$\frac{m!}{s^{m+1}}$
Exp.	e^{-at}	$\frac{1}{s+a}$
Power	$\frac{t^{m-1}e^{-at}}{(m-1)!}$	$\frac{1}{(s+a)^m}$
Sine	$\sin(at)$	$\frac{a}{s^2+a^2}$
Cosine	$\cos(at)$	$\frac{s}{s^2+a^2}$
Impulse	$\delta(t)$	1

	$u(t)$	$\hat{u}(s)$
Delay	$u(t-\tau)$	$e^{-\tau s}\hat{u}(s)$
Time-Scaling	$u(at)$	$\frac{1}{a}\hat{u}\left(\frac{s}{a}\right)$
Diff.	$u'(t)$	$s\hat{u}(s) - u(0)$
Integration	$\int_0^t u(\tau)d\tau$	$\frac{1}{s}\hat{u}(s)$
Freq. Diff.	$tu(t)$	$\frac{d}{ds}\hat{u}(s)$
Freq. Shift	$e^{at}u(t)$	$\hat{u}(s-a)$
Convolution	$\int_0^t u(\tau)v(t-\tau)d\tau$	$\hat{u}(s)\hat{v}(s)$

Tricky Example 1: $\hat{u}(s) = \frac{s}{s+1}$

$$u(t) = \delta(t) - e^{-t}$$

Tricky Example 2: $\hat{u}(s) = \frac{e^{-2s}}{s^2}$

$$u(t) = \begin{cases} 0 & t \leq 2 \\ t-2 & t \geq 2 \end{cases}$$

The Inverse Laplace Transform

An Important Case

Consider the simple case of a cosine with an exponential:

$$u(t) = e^{\sigma t} \sin \omega t$$

Approach:

- $\mathcal{L}(\sin \omega t) = \frac{\omega}{s^2+\omega^2}$
- Exponential means frequency shift: $e^{\sigma t} u(t) \mapsto \hat{u}(s-\sigma)$.

Hence the Laplace transform is

$$\hat{u}(s) = \frac{\omega}{(s-\sigma)^2 + \omega^2} = \frac{\omega}{s^2 - 2\sigma s + (\sigma^2 + \omega^2)}$$

Inverse Laplace Transform: Can we go the other direction? Calculate $u(t)$ if

$$\hat{u}(s) = \frac{1}{s^2 + bs + c}$$

Solution: Use a combination sinusoid and frequency shift. First recall that

$$\mathcal{L}(\sin \omega t) = \frac{\omega}{s^2 + \omega^2}$$

An Important Case, continued

If $\hat{u}(s) = \frac{1}{s^2+bs+c}$, the roots of the denominator are

$$s_{1,2} = -b/2 \pm \frac{1}{2}\sqrt{b^2 - 4c}$$

Hence

$$\begin{aligned}
\hat{u}(s) &= \frac{1}{(s + b/2 + \frac{1}{2}\sqrt{b^2-4c})(s + b/2 - \frac{1}{2}\sqrt{b^2-4c})} \\
&= \frac{1}{((s + b/2) + \frac{1}{2}\sqrt{b^2-4c})((s + b/2) - \frac{1}{2}\sqrt{b^2-4c})} \\
&= \frac{1}{(s + b/2)^2 - \frac{1}{4}(b^2 - 4c)}
\end{aligned}$$

Now we **REQUIRE** $4c > b^2$. Then $\hat{u}(s) = \hat{h}(s + b/2)$, where

$$\hat{h}(s) = \frac{1}{s^2 + \frac{1}{4}(4c - b^2)}.$$

Numerical Example

$$\hat{h}(s) = \frac{1}{s^2 + \frac{1}{4}(4c - b^2)}.$$

This is a simple sine function:

$$h(t) = \frac{1}{\omega} \sin(\omega t)$$

where $\omega = \sqrt{c - \frac{b^2}{4}}$

- Recall the freq.-shift property: $\mathcal{L}^{-1}(\hat{u}(s)) = \mathcal{L}^{-1}(\hat{h}(s + \frac{b}{2})) = e^{-\frac{b}{2}t} h(t)$.
- We conclude that

$$u(t) = \frac{1}{\sqrt{4c - \frac{b^2}{4}}} e^{-\frac{b}{2}t} \sin\left(\sqrt{4c - \frac{b^2}{4}}\, t\right) \qquad t \geq 0$$

Numerical Example: Let $\hat{u}(s) = \frac{1}{s^2+2s+2}$. Taking the roots, we find

$$\hat{u}(s) = \frac{1}{(s+1)^2 + 1}.$$

This is a sine function with a frequency shift of 1. Therefore

$$u(t) = e^{-t} \sin t$$

Summary

What have we learned today?

Laplace Transform of Simple Signals.

- Step, Exponentials, Ramps.
- Impulse

Properties of the Laplace Transform.

- delay, scaling
- convolution
- etc.

Next Lecture: More on Transfer Functions